

Zhifeng Gu

The Hong Kong Polytechnic University, Hong Kong
zhifeng.gu@connect.polyu.hk | <https://neal2020github.github.io/>

Education

- The Hong Kong Polytechnic University**, Hong Kong Sep 2023 – Present
- Doctor of Philosophy, Advised by Prof. Bing Wang
 - **Research Interests:** Multimodal Perception, Spatial Intelligence, World Models
- Beihang University**, Beijing, China Sep 2019 – Jun 2023
- Bachelor of Software Engineering, Advised by Prof. Lu Sheng **GPA: 3.8/4.0**

Publications

MMOne: Representing Multiple Modalities in One Scene
Zhifeng Gu, Bing Wang
International Conference on Computer Vision (ICCV 2025)

Experience

- Research on Multimodal Scene Representation** Sep 2024 – Mar 2025
- SPACE Group, HK PolyU, Advised by Prof. Bing Wang
- "MMOne: Representing Multiple Modalities in One Scene" is accepted to ICCV 2025.
 - We propose a general framework for representing multiple modalities in one scene. A modality modeling module and a multimodal decomposition mechanism are proposed to address modality conflicts.
- Research on Scene Understanding and Reconstruction** Jul 2022 – Feb 2023
- GAP Lab, CUHK-Shenzhen, Advised by Prof. Xiaoguang Han
- Use RGB videos of indoor scenes to estimate object 3D bounding boxes and reconstruct object mesh in real time.
 - GRU and attention module are utilized to better capture temporal relationship to improve reconstruction quality. Octree is adopted to speed up reconstruction process.
- Research on Object Pose Estimation and Tracking** Jun 2022 – Jun 2023
- VCG Lab, Beihang University, Advised by Prof. Lu Sheng
- Focus on human-object interaction scenes, where existing methods perform badly due to hand occlusion.
 - Adopt a motion-centric paradigm, where we explicitly model the object motion between successive frames and refine poses through shape completion and attention mechanism using previous frames.

Selected Honors and Awards

- Outstanding Graduate in Beihang University, 2023

Selected Skills

- Python, PyTorch, C/C++ , Java, etc.